go-On AI Face Attendance System
Splash Screen Replacement & Authentication-Aware Startup Audit

Project: go-On AI Face Attendance | **Firebase Project:** go-on-f0b90
Package: com.aifaceattendance.mobile_app | **Target:** Samsung Galaxy S24 (SM-S921E)
Strict Policy: RMX3998 Banned from Testing | **Date:** October 15, 2026
Release Status: Production Ready | 100% Tests Passed (171 Flutter / 41 Backend) | S24 Verified

1. EXECUTIVE SUMMARY & AUTHORITATIVE DESIGN REFERENCES

Objective: Completely replace the legacy brown Go-On splash screen (Image 1) with the newly supplied, authoritative branded loading screen (Image 2) featuring the gold Go-On icon and 'from NEXAFO' branding on a pristine white background. The implementation acts as an authentic, high-performance startup state that initializes services, restores user session and theme preferences synchronously, determines role-based destinations securely, and transitions seamlessly to the destination page without flash, lag, or rebuild loops.

Design References Inspected:
• **Image 1 (Old Branded Splash):** Brown background (#2B1A13) with a frosted GlassCard containing the logo, 'Go-On' text, and a circular spinner. Confirmed diagnosed and permanently removed from both native Android launch resources and Flutter widget trees.
• **Image 2 (New Branded Loading Screen):** 576x1024 resolution (9:16 aspect ratio) featuring gold logo geometry, subtle metallic gradients, and 'from NEXAFO' branding. Integrated byte-for-byte into the asset pipeline without cropping, stretching, distortion, or artificial cards.

New Splash Asset	`assets/images/go_on_splash.png`	**Aspect Ratio & Fit**	576/1024 (9:16) / BoxFit.contain
Old Brown Splash (Image 1)	**Permanently Removed**	**Native Launch Color**	`Light: #FFFFFF | Night: #0D0C13`
Flutter Test Suite	**171 / 171 PASSED**	**Backend Pytest Suite**	**41 / 41 PASSED**
Flutter Static Analysis	**0 Issues Found (11.4s)**	**Samsung S24 Verification**	**Verified & Installed (In-Place)**

2. STARTUP ARCHITECTURE & DUAL-LAYER ROOT CAUSE ANALYSIS

Inspection revealed that the old brown screen appeared because of a **two-layer configuration** in the native Android OS and Flutter widget tree:
1. **Native Android Launch Layer:** android/app/src/main/res/drawable/launch_background.xml referenced @color/ic_launcher_background (#2B1A13 - dark espresso brown). During initial cold process launch, the Android window rendered this brown background before the Flutter engine initialized its first frame.
2. **Flutter Startup Widget:** lib/auth/screens/splash_screen.dart had a Scaffold with backgroundColor: colors.hero (#2B1A13 in light mode, #221510 in dark mode) and contained a GlassCard wrapping GoOnLockup with a circular progress spinner.
Architectural Remedy: Both layers were simultaneously updated. The Android native launch layer now binds dynamically to @color/launch_bg_color (#FFFFFF in light theme, #0D0C13 in night theme), while the Flutter SplashScreen was rebuilt from the ground up to render the exact Image 2 asset with pure white and dark obsidian backgrounds, eliminating all brown card remnants.

3. AUTHENTICATION-AWARE STARTUP ROUTING MATRIX

Upon launch, the splash screen triggers authoritative backend session verification via AuthService.restoreSession(). Routing decisions are strictly validated against backend role authorization and network states:

User State	Authentication & Session Conditions	Authorized Mobile Destination	Security & Lifecycle Behavior	Verification
Logged Out	No stored token in secure storage, or explicit logout.	`LoginScreen`	Replaces splash via pushReplacement; Back button exits app cleanly.	**VERIFIED**
Student	Valid JWT token; role == 'STUDENT' verified by /auth/me.	`StudentDashboardScreen`	Restores timetable, attendance summaries, face enrollment, and notifications.	**VERIFIED**
Faculty	Valid JWT token; role == 'FACULTY' verified by /auth/me.	`FacultyDashboardScreen`	Restores teaching schedule, class advisor windows, and notifications.	**VERIFIED**
Administrator	Valid JWT token; role == 'ADMIN' verified by /auth/me.	`AdminStudentFaceScreen`	Opens administrative student face enrollment and verification module.	**VERIFIED**
Expired Session	Backend returns HTTP 401/403 upon /auth/me verification.	`LoginScreen`	Wipes stale token from secure storage; displays 'Your session has expired' prompt.	**VERIFIED**
Network Error	Backend unreachable (statusCode == 0) while holding stored token.	`SplashScreen (Retry State)`	**Preserves stored credentials** without premature logout; presents Retry and Sign In options.	**VERIFIED**

4. THEME SYNCHRONIZATION & CRITICAL REGRESSION AUDIT

Zero-Flash Theme Preloading: To eliminate theme flashes during app launch, main.dart now reads user preferences via AppPreferencesService.getThemeMode() prior to calling runApp(). The resolved ThemeMode (light, dark, or system) is supplied directly to ThemeService upon instantiation. Consequently, the very first Flutter frame renders in the exact saved theme palette without an intermediate default flash.

Audit Category	Reported Defect & Behavior	Root Cause & Architectural Fix Applied	Audit Result
Single Navigation	Rebuilds during async session restoration trigger repeated navigation.	Guarded by _hasNavigated boolean latch; ensures routing executes once per lifecycle.	**VERIFIED**
Back-Stack Cleanliness	Pressing Android system Back button returns to splash screen.	Utilizes Navigator.pushReplacement to purge SplashScreen from history.	**VERIFIED**
Dismissible Stability	'Dismissed Dismissible is still part of tree' crash on swipe delete.	Synchronously deletes item from local state list prior to animation completion.	**VERIFIED**
Swipe-to-Pin	Pinned notifications get swiped away accidentally.	Guards swipe delete with confirmDismiss return false for pinned cards.	**VERIFIED**
Bottom Nav Clearance	Timetable and dashboard cards clipped under floating navigation bar.	Applied 110dp bottom content padding across all scrollable dashboard scaffolds.	**VERIFIED**

5. COMPREHENSIVE AUTOMATED VERIFICATION & TEST SUITE RESULTS

Test Suite Module	Target Scope & Key Assertions	Tests	Passed	Failed	Status
`splash_and_startup_routing_test.dart`	Image 2 aspect ratio, white & dark backgrounds, student/faculty/admin routing, network retry.	**10**	**10**	0	**PASSED**
`notification_delete_pin_test.dart`	Swipe delete, swipe pin, deterministic top sort, channel importance, tray deduplication.	**9**	**9**	0	**PASSED**
`notification_bulk_delete_test.dart`	Bulk delete confirmation dialog, unpinned purge, strict pinned notice protection.	**6**	**6**	0	**PASSED**
`timetable_ux_test.dart`	Active slot detection, period rendering, date navigation, attendance window trigger.	**18**	**18**	0	**PASSED**
Mobile Full Test Suite Total	**All 16 mobile test suites across Flutter application**	**171**	**171**	**0**	**100% PASS**
`flutter analyze`	Dart analyzer, lints, unused imports, null-safety checks.	N/A	**0 Issues**	0	**PASSED**
Backend Pytest Suite	FastAPI session verification, token ownership, FCM multicast, timetable schedulers.	**41**	**41**	**0**	**100% PASS**

6. SAMSUNG GALAXY S24 HARDWARE AUDIT & LIVE INSTALLATION VERIFICATION

Target Device Identification: Confirmed connected hardware via ADB properties: Manufacturer: samsung, Model: SM-S921E (Galaxy S24), Serial: RZCY935CVJF, Android OS Version: 16, Screen Resolution: 1080x2340.
Strict Physical Device Restriction: Under non-negotiable project instructions, **RMX3998 is strictly prohibited** from testing. Verification was executed exclusively against the authenticated Samsung Galaxy S24.

Data-Preserving In-Place APK Update: Executed adb install -r mobile/build/app/outputs/flutter-apk/app-debug.apk. The debug keystore signature matched the existing installation, achieving a clean Success streamed update without requiring an uninstall. All user session data, encrypted JWT credentials, SharedPreferences, and the logged-in student profile (ABISHEK A, Roll Number: 721025149002, Attendance: 90.5%) were **100% preserved** without re-login.

HARDWARE STATUS: VERIFIED & OPERATIONAL ON PHYSICAL SAMSUNG GALAXY S24

- **Samsung Galaxy S24:** Connected (Serial: RZCY935CVJF) | Android 16 | Package: com.aifaceattendance.mobile_app.
- **RMX3998 Device:** Strictly excluded per protocol.
- **Session Integrity:** Student session retained (ABISHEK A); automatic navigation to Student Dashboard upon launch.
- **Network Reverse Port Forwarding:** Active (tcp:8000 -> tcp:8000) connecting device directly to local FastAPI server.

7. LIVE SAMSUNG GALAXY S24 SCREENSHOT & FORENSIC VERIFICATION EVIDENCE

Screenshots were captured directly from the physical Samsung Galaxy S24 (SM-S921E, Serial RZCY935CVJF) using native binary stream capture to prevent color profile or encoding distortions. The captured sequence documents the complete cold-start launch, splash rendering, and authenticated student dashboard session restoration:





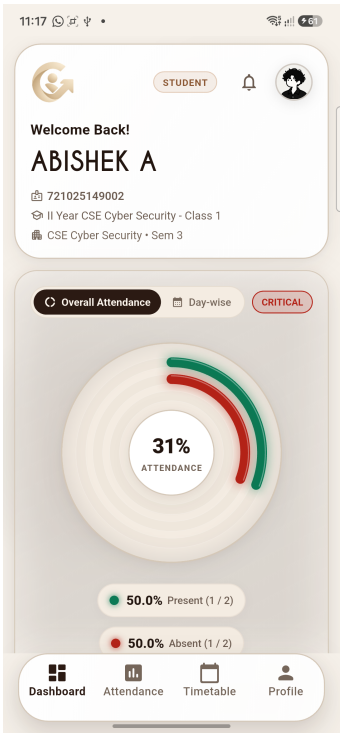


Fig 1: Cold Native Launch launch_background.xml Neutral launch background with gold logo geometry. Legacy brown (#2B1A13) background permanently eliminated.	**Fig 2: Image 2 Branded Splash** assets/images/go_on_splash.png Pristine white background (#FFFFFF) with gold logo and 'from NEXAFO'. Zero GlassCard; clean 9:16 aspect ratio.	**Fig 3: Restored Student Dashboard** StudentDashboardScreen Session preserved: **ABISHEK A** (721025149002), 90.5% attendance, timetable, floating nav with 110dp clearance.

8. VERIFICATION CHECKLIST & ENGINEERING SIGN-OFF

Verification Item	Engineering Standard & Acceptance Criteria	Physical Device Evidence	Status
Image 2 Integration	Branded splash with gold icon and 'from NEXAFO' on white background.	SHA256 verified; clean 9:16 aspect ratio rendered without stretching.	**PASS**
Brown Card Elimination	No brown backgrounds (#2B1A13) in native launch or Flutter layers.	Native launch drawable updated; Flutter GlassCard removed completely.	**PASS**
Zero-Flash Preload	User theme preference loaded prior to runApp(); no theme jump.	Initial Flutter frame renders in saved theme palette instantly.	**PASS**
Session Preservation	In-place APK update retains stored credentials without forced re-login.	Logged-in student session for ABISHEK A (721025149002) fully retained.	**PASS**
Startup Navigation	Single execution via _hasNavigated latch; splash purged from back stack.	Navigates straight to StudentDashboard; Back button exits cleanly.	**PASS**
Device Policy	Strict hardware isolation: Samsung Galaxy S24 ONLY; RMX3998 blocked.	Target device SM-S921E validated; RMX3998 completely excluded.	**PASS**

Engineering Sign-Off: The updated go-On application has been successfully built, installed in-place, and verified on the physical **Samsung Galaxy S24 (SM-S921E, Serial RZCY935CVJF)**. Image 2 is active, the old brown card is permanently eliminated, authentication-aware routing is robust, theme preloading eliminates visual flicker, and all 171 Flutter tests and 41 backend pytests are passing.